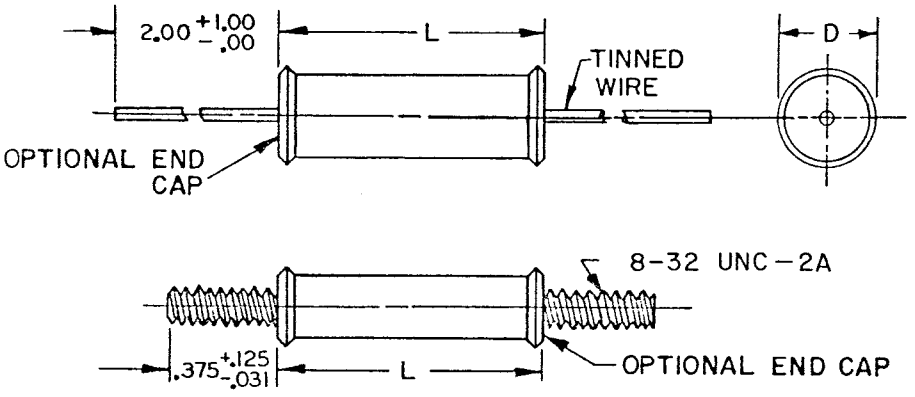
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC (HERMETICALLY SEALED IN CERAMIC OR GLASS CASES) STYLE CQ20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-19978.



INCHES	mm
.001	0.03
.004	0.10
.031	0.79
.032	0.81
.040	1.02
.125	3.18
.375	9.53
.531	13.49
.625	15.88
.750	19.05
1.00	25.4
2.00	50.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only.
(C) 3. Wire leads shall be tinned solid wire, .032 (No. 20 AWG) for cases .531 and .625 in diameter, and .040 (No. 18 AWG) for cases .75 and above in diameter. Tolerance on all lead wire diameters shall be +.004, -.001.
4. Mounting hardware consisting of two lockwashers and four nuts shall be supplied with each capacitor with threaded-stud terminals.
(C) 5. These capacitors are not intended to be supported by their leads. These capacitors shall be supported with a supplementary means of mounting, such as tangential retainer or a wrap-around band. The supporting device will not be supplied with the capacitor.

FIGURE 1. Style CQ20 capacitors.

(C) denotes changes

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REQUIREMENTS:

Design and construction:

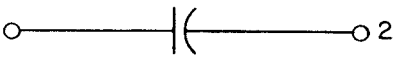
Ⓒ Dimensions and configuration: See figure 1 and table I.

Construction: Extended foil or laid-in tab.

Case material: Ceramic or glass.

Dielectric material: Paper and polyethylene terephthalate (K).

Ⓒ Terminals: See table I.

Circuit diagram: 1  2

Ⓒ Rated voltage: See table I.

Rated temperature: -65°C to +125°C.

Ⓒ Capacitance (Cap.)(nom): See table I.

Ⓒ Capacitance tolerance: See table I.

Dissipation factor (DF)(max): 1.0 percent.

Ⓒ TABLE I. Style CQ20 capacitors.

Part number 1/	DC voltage rating (see figure 2)	Capacitance	Capacitance tolerance	Terminal identification	Case dimensions 2/	
					L ±.062	D ±.031
	Volts	µF			Inches	Inches
CQ20A1KG102-1	1,000	0.00010	J, K	A	1.188	.531
CQ20A1KG152-1	1,000	0.0015	J, K	A	1.188	.531
CQ20A1KG222-1	1,000	0.0022	J, K	A	1.188	.531
CQ20A1KG332-1	1,000	0.0033	J, K	A	1.188	.531
CQ20A1KG472-1	1,000	0.0047	J, K	A	1.188	.531
CQ20A1KG682-1	1,000	0.0068	J, K	A	1.188	.531
CQ20A1KG103-1	1,000	0.010	J, K	A	1.188	.531
CQ20A1KG153-1	1,000	0.015	J, K	A	1.188	.531
CQ20A1KG223-1	1,000	0.022	J, K	A	1.562	.531
CQ20-1KG333-1	1,000	0.033	J, K	A,C	1.562	.625
CQ20-1KG473-1	1,000	0.047	J, K	A,C	1.750	.625
CQ20-1KG683-1	1,000	0.068	J, K	A,C	2.000	.625
CQ20-1KG104-1	1,000	0.10	J, K	A,C	2.250	.625
CQ20-1KG154-1	1,000	0.15	J, K	A,C	2.250	.750
CQ20-1KG224-1	1,000	0.22	J, K	A,C	2.500	.750
CQ20-1KG334-1	1,000	0.33	J, K	A,C	2.750	1.031
CQ20-1KG474-1	1,000	0.47	J, K	A,C	3.000	1.031
CQ20-1KG684-1	1,000	0.68	J, K	A,C	3.250	1.344
CQ20-1KG105-1	1,000	1.0	J, K	A,C	3.750	1.344
CQ20A1KJ102-1	2,000	0.0010	J, K	A	1.188	.531
CQ20A1KJ152-1	2,000	0.0015	J, K	A	1.188	.531
CQ20A1KJ222-1	2,000	0.0022	J, K	A	1.188	.531
CQ20A1KJ332-1	2,000	0.0033	J, K	A	1.188	.531
CQ20A1KJ472-1	2,000	0.0047	J, K	A	1.188	.531
CQ20A1KJ682-1	2,000	0.0068	J, K	A	1.188	.531
CQ20A1KJ103-1	2,000	0.010	J, K	A	1.562	.531
CQ20A1KJ153-1	2,000	0.015	J, K	A	1.562	.531
CQ20-1KJ223-1	2,000	0.022	J, K	A,C	1.562	.625
CQ20-1KJ333-1	2,000	0.033	J, K	A,C	1.750	.625

See footnotes at end of table.

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© TABLE I. Style CQ20 capacitors - Continued.

Part number 1/	DC voltage rating (see figure 2)	Capacitance	Capacitance tolerance	Terminal identification	Case dimensions 2/	
					L ±.062	D ±.031
	Volts	µF			Inches	Inches
CQ20-1KJ473-1	2,000	0.047	J, K	A,C	2.000	.625
CQ20-1KJ683-1	2,000	0.068	J, K	A,C	2.000	.750
CQ20-1KJ104-1	2,000	0.10	J, K	A,C	2.250	.875
CQ20-1KJ154-1	2,000	0.15	J, K	A,C	2.500	1.031
CQ20-1KJ224-1	2,000	0.22	J, K	A,C	2.750	1.031
CQ20-1KJ334-1	2,000	0.33	J, K	A,C	2.750	1.344
CQ20-1KJ474-1	2,000	0.47	J, K	A,C	3.750	1.344
CQ20-1KJ684-1	2,000	0.68	J, K	A,C	3.750	1.344
CQ20A1KL102-1	3,000	0.001	J, K	A	1.188	.531
CQ20A1KL152-1	3,000	0.0015	J, K	A	1.188	.531
CQ20A1KL222-1	3,000	0.0022	J, K	A	1.188	.531
CQ20A1KL332-1	3,000	0.0033	J, K	A	1.188	.531
CQ20A1KL472-1	3,000	0.0047	J, K	A	1.562	.531
CQ20A1KL682-1	3,000	0.0068	J, K	A	1.562	.531
CQ20-1KL103-1	3,000	0.010	J, K	A,C	1.562	.625
CQ20-1KL153-1	3,000	0.015	J, K	A,C	1.750	.625
CQ20-1KL223-1	3,000	0.022	J, K	A,C	2.000	.625
CQ20-1KL333-1	3,000	0.033	J, K	A,C	2.000	.750
CQ20-1KL473-1	3,000	0.047	J, K	A,C	2.250	.750
CQ20-1KL683-1	3,000	0.068	J, K	A,C	2.500	.875
CQ20-1KL104-1	3,000	0.10	J, K	A,C	3.250	.875
CQ20-1KL154-1	3,000	0.15	J, K	A,C	3.250	1.031
CQ20-1KL224-1	3,000	0.22	J, K	A,C	3.250	1.344
CQ20-1KL334-1	3,000	0.33	J, K	A,C	3.750	1.344
CQ20A1KN471-1	5,000	0.00047	J, K	A	1.562	.531
CQ20A1KN681-1	5,000	0.00068	J, K	A	1.562	.531
CQ20A1KN102-1	5,000	0.0010	J, K	A	1.562	.531
CQ20A1KN152-1	5,000	0.0015	J, K	A	1.562	.531
CQ20A1KN222-1	5,000	0.0022	J, K	A	1.562	.531
CQ20A1KN332-1	5,000	0.0033	J, K	A	1.562	.531
CQ20-1KN472-1	5,000	0.0047	J, K	A,C	1.562	.625
CQ20-1KN682-1	5,000	0.0068	J, K	A,C	1.750	.625
CQ20-1KN103-1	5,000	0.010	J, K	A,C	2.250	.625
CQ20-1KN153-1	5,000	0.015	J, K	A,C	2.250	.750
CQ20-1KN223-1	5,000	0.022	J, K	A,C	2.500	.875
CQ20-1KN333-1	5,000	0.033	J, K	A,C	3.000	.875
CQ20-1KN473-1	5,000	0.047	J, K	A,C	3.000	1.031
CQ20-1KN683-1	5,000	0.068	J, K	A,C	3.500	1.031
CQ20-1KN104-1	5,000	0.10	J, K	A,C	3.500	1.344
CQ20-1KN154-1	5,000	0.15	J, K	A,C	3.750	1.344
CQ20-1KR471-1	7,500	0.00047	J, K	A,C	1.562	.625
CQ20-1KR681-1	7,500	0.00068	J, K	A,C	1.562	.625
CQ20-1KR102-1	7,500	0.0010	J, K	A,C	1.750	.625
CQ20-1KR152-1	7,500	0.0015	J, K	A,C	1.750	.625
CQ20-1KR222-1	7,500	0.0022	J, K	A,C	1.750	.750
CQ20-1KR332-1	7,500	0.0033	J, K	A,C	2.250	.750
CQ20-1KR472-1	7,500	0.0047	J, K	A,C	2.250	.750
CQ20-1KR682-1	7,500	0.0068	J, K	A,C	2.750	.875
CQ20-1KR103-1	7,500	0.010	J, K	A,C	2.750	.875
CQ20-1KR153-1	7,500	0.015	J, K	A,C	2.750	1.031
CQ20-1KR223-1	7,500	0.022	J, K	A,C	2.750	1.344
CQ20-1KR333-1	7,500	0.033	J, K	A,C	2.750	1.344
CQ20-1KR473-1	7,500	0.047	J, K	A,C	3.500	1.344
CQ20-1KR683-1	7,500	0.068	J, K	A,C	4.250	1.344
CQ20-1KS471-1	10,000	0.00047	J, K	A,C	2.250	.625
CQ20-1KS681-1	10,000	0.00068	J, K	A,C	2.250	.625
CQ20-1KS102-1	10,000	0.0010	J, K	A,C	2.250	.750
CQ20-1KS152-1	10,000	0.0015	J, K	A,C	2.250	.875
CQ20-1KS222-1	10,000	0.0022	J, K	A,C	2.750	.875

See footnotes at end of table.

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© TABLE I. Style CQ20 capacitors - Continued.

Part number 1/	DC voltage rating (see figure 2)	Capacitance	Capacitance tolerance	Terminal identification	Case dimensions 2/	
					L ±.062	D ±.031
	Volts	µF			Inches	Inches
CQ20-1KS332-1	10,000	0.0033	J, K	A,C	2.750	.875
CQ20-1KS472-1	10,000	0.0047	J, K	A,C	2.750	1.031
CQ20-1KS682-1	10,000	0.0068	J, K	A,C	2.750	1.344
CQ20-1KS103-1	10,000	0.010	J, K	A,C	2.750	1.344
CQ20-1KS153-1	10,000	0.015	J, K	A,C	3.250	1.344
CQ20-1KS223-1	10,000	0.022	J, K	A,C	3.750	1.344
CQ20-1KS333-1	10,000	0.033	J, K	A,C	4.750	1.344
CQ20-1KS393-1	10,000	0.039	J, K	A,C	5.250	1.344
CQ20-1KT471-1	12,500	0.00047	J, K	A,C	2.750	.750
CQ20-1KT681-1	12,500	0.00068	J, K	A,C	2.750	.750
CQ20-1KT102-1	12,500	0.0010	J, K	A,C	2.750	.875
CQ20-1KT152-1	12,500	0.0015	J, K	A,C	3.250	.875
CQ20-1KT222-1	12,500	0.0022	J, K	A,C	3.250	1.031
CQ20-1KT332-1	12,500	0.0033	J, K	A,C	3.750	1.031
CQ20-1KT472-1	12,500	0.0047	J, K	A,C	3.750	1.031
CQ20-1KT682-1	12,500	0.0068	J, K	A,C	3.250	1.344
CQ20-1KT103-1	12,500	0.010	J, K	A,C	3.750	1.344
CQ20-1KT153-1	12,500	0.015	J, K	A,C	4.750	1.344
CQ20-1KT223-1	12,500	0.022	J, K	A,C	5.250	1.344
CQ20-1KU471-1	15,000	0.00047	J, K	A,C	2.750	.750
CQ20-1KU681-1	15,000	0.00068	J, K	A,C	2.750	.875
CQ20-1KU102-1	15,000	0.0010	J, K	A,C	3.250	.875
CQ20-1KU152-1	15,000	0.0015	J, K	A,C	3.250	.875
CQ20-1KU222-1	15,000	0.0022	J, K	A,C	3.250	1.031
CQ20-1KU332-1	15,000	0.0033	J, K	A,C	3.250	1.344
CQ20-1KU472-1	15,000	0.0047	J, K	A,C	3.250	1.344
CQ20-1KU682-1	15,000	0.0068	J, K	A,C	3.750	1.344
CQ20-1KU103-1	15,000	0.010	J, K	A,C	4.250	1.344
CQ20-1KU153-1	15,000	0.015	J, K	A,C	5.250	1.344

© 1/ Complete part number shall include symbols to indicate terminal and capacitance tolerances.

© 2/ See table III for metric equivalents.

Seal: In accordance with method 112 of MIL-STD-202.

Test condition letter: A.

Dielectric withstanding voltage (DWV):

Sleeving: In accordance with MIL-C-19978.

Barometric pressure: In accordance with MIL-C-19978.

Test points:

Circuit diagram 1: Between terminals and case.

Insulation resistance (IR):

Sleeving: In accordance with MIL-C-19978.

© Terminal to terminal: See table II.

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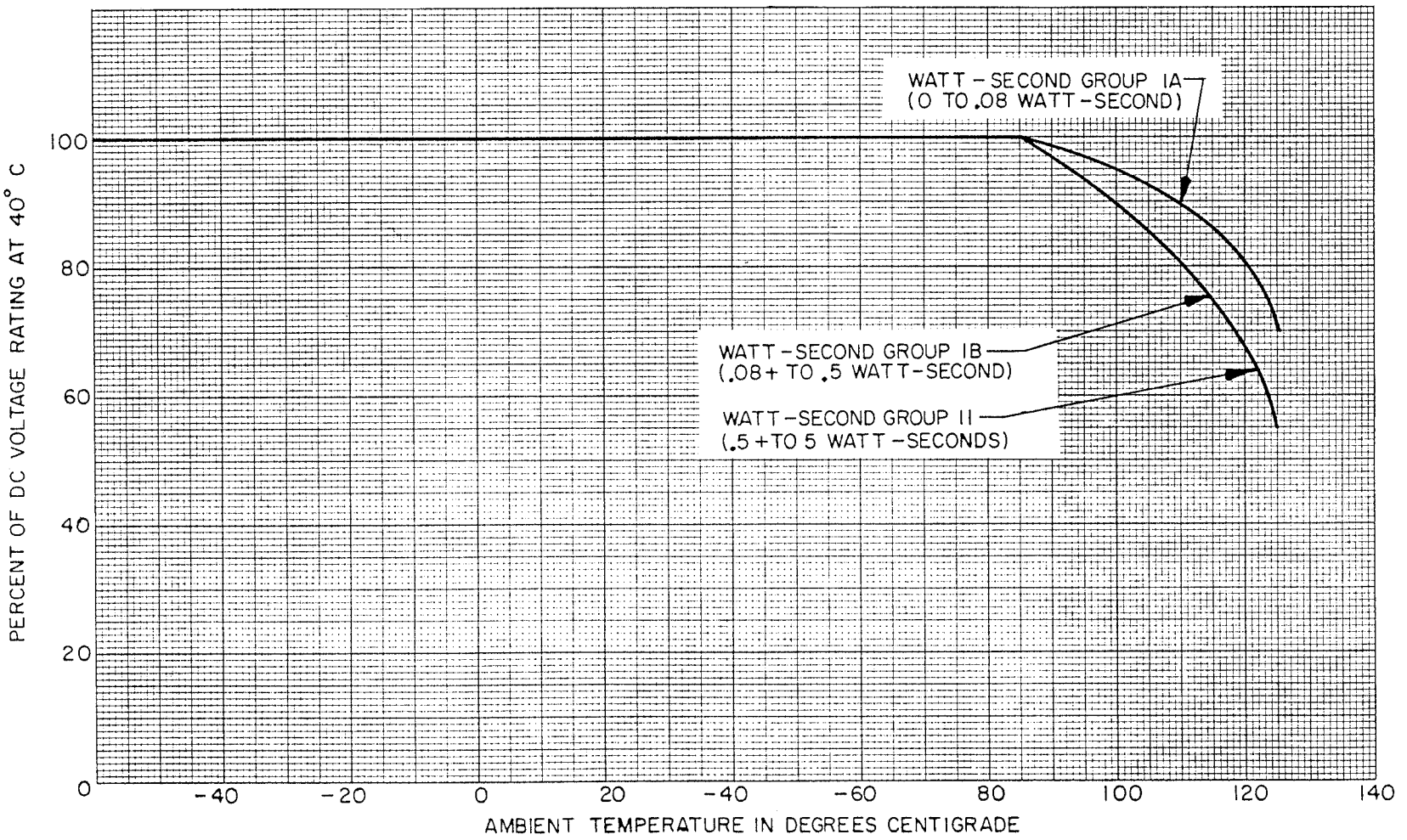


FIGURE 2. Voltage derating for high ambient test temperature (characteristic K).

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© TABLE II. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
Characteristic K	At 25°C
0 to 0.6 microfarad	25,000 megohms
Greater than 0.6 microfarad	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad	250 megohms
Greater than 0.08 microfarad	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Vibration: In accordance with method 201 of MIL-STD-202.

Low frequency:

Direction and duration of motion: Four hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: Not applicable.

Immersion:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978 for wire leads only.

Terminal strength: In accordance with method 211 of MIL-STD-202.

Test condition letter: D (wire leads).

Test condition letter: A and E (threaded stud and nuts).

Moisture resistance:

DWV:

Sleeving: In accordance with MIL-C-19978.

IR:

Sleeving: In accordance with MIL-C-19978.

Stability at low and high temperatures:

Low temperature:

Test temperature: -65°C +0°C, -5°C.

Capacitance change (max): -10 percent.

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High temperature:

Test temperature: +125°C +0°C, -5°C.

Capacitance change (max): +10 percent.

Life:

Capacitance change (max): ±5 percent.

Resistance to soldering heat: In accordance with MIL-C-19978.

TABLE III. Millimeter equivalents of decimal inches.

Inches	mm
.031	0.79
.062	1.57
.531	13.49
.625	15.88
.750	19.05
.875	22.23
1.031	26.19
1.188	30.18
1.344	34.14
1.562	39.67
1.750	44.45
2.000	50.80
2.250	57.15
2.500	63.50
2.750	69.85
3.000	76.20
3.250	82.55
3.500	88.90
3.750	95.25
4.250	107.95
4.750	114.30
5.250	133.35

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, TD
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC, OS, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5910-1560)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-19978/2C	CAPACITORS,FIXED,PLASTIC(OR PAPER-PLASTIC) DI-ELECTRIC (HERMETICALLY SEALED IN CERAMIC OR GLASS CASES STYLE CQ20

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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